Project Plan (Project Debriefing)

C. Stijlaart

Version 1.1, 15-10-2023

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1. Document history

Table 1. Document History

Version	Date	Status	Author	Description	Remarks
1.1	15-10-2023	First Version	Casey Stijlaart	Complete first version	-

2. Project Assignment

2.1. Context

The project's revolves around the creation of an embedded system that processes environmental data and transforms it into musical compositions using algorithms and IoT technology. This interests originates from the desire to gain a deeper understanding of algorithms and explore IoT connectivity further. It represents a shift towards a more research-oriented and self-directed learning experience, in contrast to previous projects with predefined constraints.

2.1.1. Problem analysis

A common challenge that many individuals encounter is the understanding of text-based data. Text-based data tends to be flat and lacking the necessary context for a good understanding. Consequently, people often struggle to assess the quality of their environment, be it in terms of temperature, sound levels, or other factors.

2.1.2. Problem definition

The problem to be addressed here is the difficulty people face when attempting to understand environmental conditions represented in text-based data. Such data is often insufficient in providing the context needed for an accurate assessment of the environment's quality. The result is that individuals may lack a clear understanding of whether their climate conditions are optimal or suboptimal, making it challenging to respond to potential issues or improvements effectively.

2.2. Project Goal

The goal is to successfully design and implement an embedded system that effectively utilizes environmental data to create musical compositions via algorithmic processes. The musical composition should give a clear indication to the human on the environmental status. Allowing for data to be informative in a non-text-based manner,

2.3. Assignment

In this personal project, the assignment is to design and implement an embedded system that utilizes environmental data, such as temperature and sound levels, to generate musical output through the use of algorithms and IoT connectivity. The primary objective is to expand skills and knowledge in algorithmic music generation and IoT integration, while embracing a more research-based approach.

2.4. Finished Products

The finalized product exists out of multiple parts, documentation, software and prototype setup. Within the following tables, [Documentation], [Prototype] and [Final Version], the final deliverables are given.

Table 2. Documentation

Documentation
Project Plan
System Design Document
Technical Document

Table 3. ProtoType

Prototype
Database with retrieval and storing data
Communication between components
Data processing for musical output

Table 4. Final Version

Final Version
Working prototype

2.5. Research Questions

In order to reach the deliverable, a comprehensive exploration of various facets must be explored. These diverse facets each play an important role in determining the functioning of the system. To delve into these crucial aspects, research questions are created. Each research question covering their own set of facets.

Main question:

How can environmental data be harnessed to produce meaningful musical output?

Sub-questions:

1. Which environmental sensor are suitable candidates?
2. What type of data structure should be used?
3. How can the retrieved data be effectively processed and utilized?
4. Which database type offers the most efficient data management?
5. What type of IoT communication protocol will be used?
6. How can musical output be generated from the collected data?
7. What methods can be used to develop a custom algorithm?
8. How can the musical output be designed to trigger specific human emotions?

2.6. Research Methods

In this project, the DOT-framework is used as primary research methodology. The DOT framework is a systematic approach that focusses on various research strategies to build system. The research strategies are Field, Lab, Library, Workshop, and Showroom research.

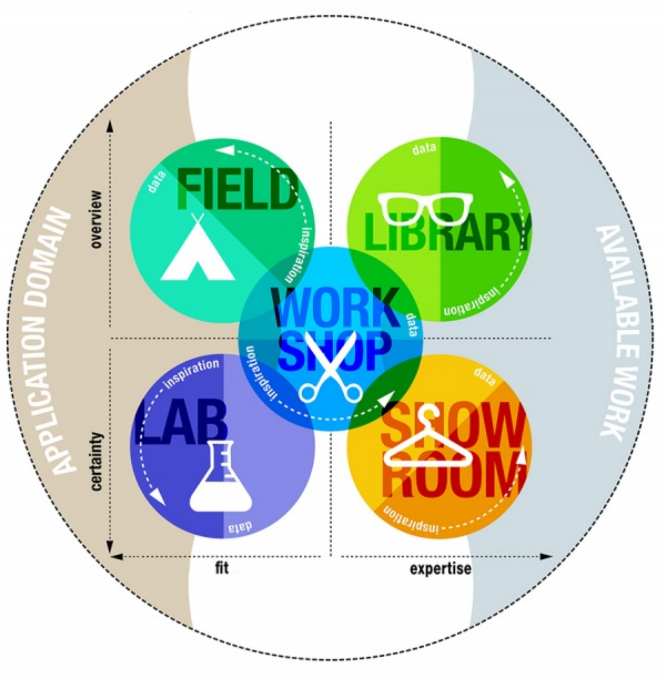

Figure 1. DOT framework

2.6.1. Research Methods x Research Questions

Each research question is met with a corresponding research strategy and technique based on the DOT framework. This approach ensures that every research question is answered effectively.

For a comprehensive view of how the DOT framework aligns with our project, consult [Research Methods and Research Questions].

Table 5. Research Methods and Research Questions

Research Question	Strategy	Technique	Steps
Which environmental sensor are suitable candidates?	Library	Literature Study, Community Research, Available Product Analysis	1. Conduct an extensive literature study on environmental sensors. 2. Engage in community research to gather insights from experts and users. 3. Analyze available environmental sensor products in the market.

Research Question	Strategy	Technique	Steps
What type of data structure should be used?	Library, Field, Workshop	Literature Study, Document Analysis, Prototyping	1. Perform an in-depth literature study on data structures suitable for embedded systems. 2. Analyze documents and research papers related to data structures. 3. Create prototypes to evaluate the practicality of data structures in the field.
How can the retrieved data be effectively processed and utilized?	Library, Lab, Workshop	Literature Study, Data Analytics, Prototyping	1. Review relevant literature on data processing methods. 2. Utilize data analytics techniques to process and analyze collected data. 3. Develop and test prototypes for effective data utilization.
Which database type offers the most efficient data management?	Library, Lab, Workshop	Literature Study, Community Research, Usability Testing, Prototyping	1. Research and compare various database types through literature study. 2. Gather insights from the community to understand their experiences with database management. 3. Conduct usability testing to assess database efficiency. 4. Develop database prototypes for evaluation.
What type of IoT communication protocol will be used?	Library	Literature Study, Community Research	1. Conduct a literature study to explore IoT communication protocols. 2. Engage with the community to understand the practicality and performance of different protocols.

Research Question	Strategy	Technique	Steps
How can musical output be generated from the collected data?	Library, Workshop	Literature Study, Community Research, Usability Testing	1. Study relevant literature on algorithms for music generation from data. 2. Seek input from the community regarding their preferences and experiences. 3. Conduct usability testing to refine and improve the music generation process.
What methods can be used to develop a custom algorithm?	Library, Lab, Workshop	Community Research, Literature Study, Prototyping, Component Test, Data Analytics	1. Engage in community research to identify preferred algorithm development methods. 2. Conduct a comprehensive literature study on algorithm development. 3. Create prototypes and perform component testing. 4. Utilize data analytics to refine custom algorithms.
How can the musical output be designed to trigger specific human emotions?	Showroom, Library	Product Review, Peer Review, Literature Study	1. Showcase the musical output and gather feedback from users for product review. 2. Seek peer reviews and conduct literature study to enhance the music design based on emotional triggers.

3. Approach and Planning

3.1. Approach

Throughout the project, I am following the scrum methodology. Sprints are utilized to track progress and plan for the next steps. Each sprint typically spans a month, with adjustments made for holidays. At the end of a sprint, a sprint demo is conducted to showcase the work and accomplishments achieved during that sprint.

3.1.1. Test Approach

To ensure a seamless software implementation, I have integrated testing into the project. Testing takes different forms, each focusing on different aspects of the project. The primary testing methods employed in this project are unit testing and integration testing, which help ensure the quality and functionality of the software.

3.2. Breakdown of the Project

The project is structured into four distinct phases, each with its unique purpose. These phases serve as a roadmap, providing clear guidance and maintaining a well-defined trajectory throughout the project.

Analysis and problem definition

The initial phase holds the most importance, as it lays the groundwork for the entire project. Here, the emphasis is on defining the problem, exploring solutions, and setting up a plan. This plan acts as a guidance throughout the project, ensuring alignment with the requirements.

Design

In the second phase, attention shifts towards creating the system’s design. Meaning that the overall system architecture is created, aided by various diagrams, including flow charts and state diagrams.

Realization

In the third stage, the design and overall plans are put into a 'real' system. All the knowledge acquired is put to the test, and a continuous cycle of software development and testing ensues. Once the envisioned result is achieved, the project advances to the next phase.

Evaluation

The final phase centers on presenting and assessing everything created and learned in the preceding stages. This includes a demonstration of the final product, followed by an in-depth evaluation to ensure it meets the desired standards.

3.3. Timeline

The project is guided by multiple timelines, each playing a important role in its progress. These timelines provide structure, to ensure that the project remains well-organized and on track.

3.3.1. Overall Timeline

The project's overall timeline, depicted in [Project Timeline], offers a high-level overview of project phases and significant events.

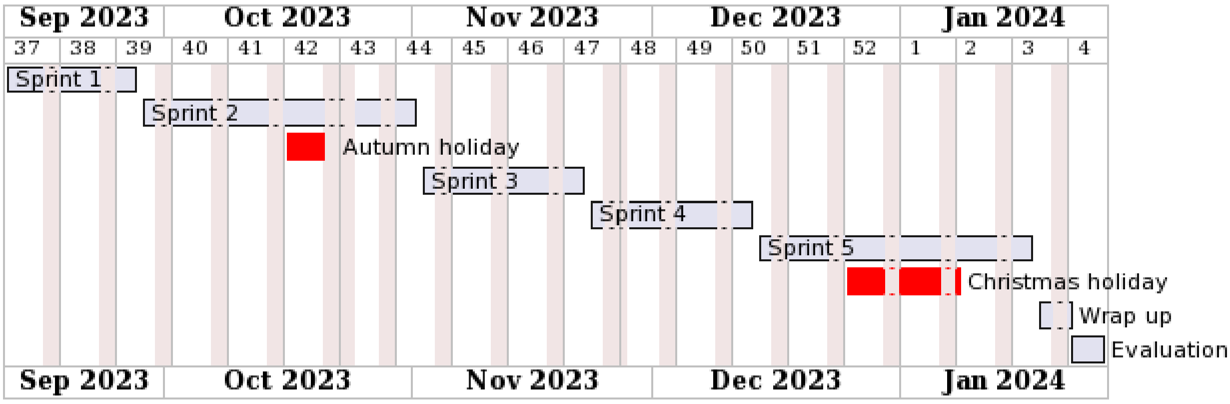

Figure 2. Project Timeline

3.3.2. Tasks Timeline

To provide a clear view of the project's approach and an estimate of its phase durations, a detailed table, [Expected Tasks and Phasing Duration], outlines the tasks executed during each sprint. For a more sketched-out week-by-week view of the phases and sprints, please refer to the timeline depicted in [Phasing Timeline].

Table 6. Expected Tasks and phasing duration

Sprint	Phase	Tasks
1	Analysis and Problem Definition	Research the problem, the solutions, and its requirements. Next to this, create the concrete plan.
2	Design	Create the design.
3	Realization	Implementation and testing.
4	Realization	Implementation and testing.
5	Realization / Evaluation	Implementation, testing and final realization.

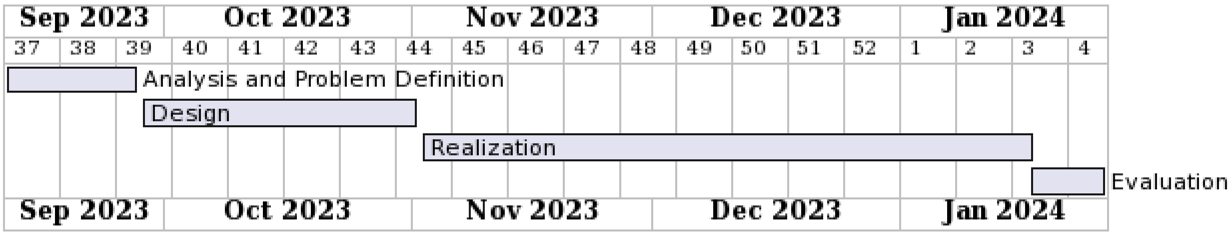

Figure 3. Phasing Timeline

4. Project Organization

4.1. Team Members

4.1.1. Project group members

Table 7. Team Members

Name	**E-mail**
Casey Stijlaart	c.stijlaart@student.fontys.nl

4.1.2. Coaches

Table 8. Coaches

Name	E-mail	Remarks
Cees van Tilborg	c.vantilborg-ab@fontys.nl	1st coach. General mentor & first point of contact. Semester Coach.

4.2. Communication

The main communication is with the coach, Cees van Tilborg. Teams and email are the main way of communication. To ensure the coach is update with my progress and overall project status, sprint meetings are held every month.

4.3. Test environment

Within the project there is made use of various approach to verify the system and its software. There are two approaches that will be used. These approaches are integration testing and unit testing. Both tests a different aspect of the system. Where unit testing focusses more on the actual software side, the integration testing focuses more on testing the various components together.

Integration testing is done each time components are added. This to ensure the system still operates in the expected manner.

When it comes to unit testing, it allows for a continuous verification of the software. Individual parts are tested in an more isolated manner, ensuring that each individual parts do not have any hidden bugs. Besides that, when adding new code is added, unit tests will ensure that the software will still work accordingly.

4.4. Configuration Management

To ensure proper version control, GitLab is used for this project. All project files are stored on GitLab, with a specific GitLab environment, InnovationLab, provided by Fontys. Within Git, I utilize "issues" to manage my tasks. Each issue corresponds to a small task within the project. I can easily divide and keep track of my tasks using these issues. To ensure the safety and integrity of my code, I create a branch for each issue and work on this branch exclusively for the duration of the task.

Once I’ve completed an issue, I can merge the branch to the main branch.

5. Risk Analysis

5.1. Risks and Fallback Activities

Every project carries risks, each with its unique implications and potential solutions. To gain insight into the potential risks, their resolutions, and prevention strategies, please refer to the following details in [Risk and Contingency Plan].

Table 9. Risk and Fallback Activities

Risk	Prevention Activities in the Plan	Contingency Measures
Running Out of Time.	A well-structured plan with built-in time buffers can allow for time-related risks. Ensuring extra planned in time for each project phase allows for flexibility when unexpected issues occur.	Discuss the matter with project coach, and assess the situation to determine the way forward.
Long-term Sickness.	Clear communication with the project coach to ensure the coach is up-to-date, and has a clear idea on what is going on.	Discuss possible actions that could be taken and solution to be found.